

Payanshi Jain

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Bengaluru, India

Computer Science undergraduate at **Scaler School of Technology (BITS Pilani)** focused on building robust web architectures, high-performance spatial CSS interfaces, and distributed systems tested against deliberate failure conditions.

EDUCATION

Scaler School of Technology (Degree awarded by BITS Pilani)

2026 – 2029

B.Tech in Computer Science · Bengaluru, India

Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java), Operating Systems, Web Systems Architecture, Database Internals.

PROJECTS

Kin — Shared Medication Schedule for Family Caregivers

2026

[React](#) · [Node.js](#) · [MongoDB](#) · [WebSockets](#) · [Vitest](#)

- Real-time care coordination web application built so two family caregivers never administer duplicate doses. Real-time updates push through WebSockets the instant a dose is recorded.
- Implemented atomic `findOneAndUpdate` arbitration to resolve simultaneous tap collisions cleanly. Verified against 5 concurrent mobile client taps with zero double-dose race conditions.
- Architected accessible WCAG AAA compliant interface with high-contrast modes and fluid typography.

Flexbox Quest — Interactive Game Layout Engine

2026

[HTML5 Semantic](#) · [CSS3 Flexbox](#) · [Responsive Design](#) · [Spatial Mechanics](#)

- Designed and developed a game-inspired spatial interface showcasing advanced CSS Flexbox mechanics, multi-axis alignments, responsive wrapping, and interactive HUD status boards.
- Engineered zero-runtime layout calculations using pure modern CSS properties (`justify-content: space-between`, `align-items: center`, dynamic clamping).

Stow — Embedded Key-Value Storage Tested by Simulated Power Loss

2026

[Go](#) · [Write-Ahead Log \(WAL\)](#) · [Memtable](#) · [Crash Injection](#) · [Virtual FS](#)

- Engineered an embedded storage engine in Go with Write-Ahead Logging and an instrumented mock VFS that injects simulated crashes across 43 durability-relevant disk write points.
- Tested against clean crashes and three torn-write variants (~170 recovery scenarios per test run) to guarantee atomic commit durability without data corruption.

Student Portal — Academic Records & Course Registration System

2026

[HTML5 Semantic](#) · [Advanced CSS](#) · [Form Validation](#) · [Accessible Tables](#)

- Built an academic dashboard featuring student profiles, responsive tabular schedule layouts, and structured course registration workflows with high-contrast accessibility standards.

TECHNICAL COMPETENCIES

Languages:	Java, JavaScript (ES6+), Go, HTML5, CSS3, SQL
Frontend & UI:	CSS Grid & Flexbox, 3D CSS Perspective, Responsive Design, Semantic HTML5, WCAG AAA a11y
Backend & Systems:	Node.js, WebSockets, MongoDB, Write-Ahead Logs (WAL), Concurrency Synchronization
Tools & Practices:	Git, GitHub, Vitest, CI/CD Automation (GitHub Actions), POSIX / Linux